Applied Thermal Engineering 107 (2016) 508–515

Contents lists available at ScienceDirect

Applied Thermal Engineering

journal homepage: www.elsevier.com/locate/apthermeng

Stabilization effects of perforated plates on the combustion instability in a lean premixed combustor

Seungtaek Oh, Youngjun Shin, Yongmo Kim *

Department of Mechanical Engineering, Hanyang University, Sungdong-Ku, Seoul 133-791, Republic of Korea

ARTICLE INFO

Article history:
Received 9 April 2016
Revised 21 June 2016
Accepted 21 June 2016
Available online 24 June 2016

Keywords:
Thermoacoustic instability
Flame describing function
Helmholtz solver
Attenuation characteristics
Perforated plate
Luong model

ABSTRACT

In the present study, an FDF (Flame Describing Function)-based approach is applied to numerically investigate the attenuation characteristics of a perforated plate. In order to effectively suppress the combustion instability, this study adopts an alternative approach to optimally design a passive damper using a perforated plate. We confirm that the proposed attenuation conditions of a perforated plate ensure quite stable conditions over the entire range of velocity perturbation ratios. Moreover, the present method using Luong model can be applied to all Strouhal number regimes in contrast to a conventional method using the Howe model. Numerical results were also compared with experimental data in terms of the resonant frequency and limit-cycle velocity perturbation ratio. Although slight discrepancies exist, the present FDF-based approach yields an overall agreement with the experimental results. In addition, the numerical results show that perforated plates with smaller holes or thinner plate thicknesses are more effective at stabilizing the oscillation induced by combustion.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

In order to comply with strict regulatory requirements, the gas turbine industry has typically adopted lean premixed combustors as the most promising solution [1]. However, lean premixed combustors are quite susceptible to combustion instability, which leads to serious failure of the system [2]. Therefore, the combustion instability must be technically resolved prior to system operation. Numerical studies have been carried out in attempts to understand the driving mechanism of the combustion instability as well as to optimally suppress the thermoacoustic oscillations. These numerical models are largely classified by the self-excited mode strategy and the forced mode strategy [1].

In the self-excited mode strategy, CFD calculations need to be carried out until small fluctuations are developed with a certain resonant frequency. Thus, external forcing is not required because the flow field has its own instability modes. In the past, unsteady RANS (URANS) based turbulent combustion models [1,3–7] have been employed to numerically analyze the combustion instability; however, these approaches are limited in their ability to realistically capture the physical processes involved in the combustion instability. To resolve these shortcomings, the large eddy simulation (LES) based turbulent combustion model has frequently been applied to predict unstable combustion dynamics [1,8–13]. It is widely recognized that the LES-based turbulent combustion model is the ultimate design tool for identifying the optimal stable conditions of real combustors. However, the prediction capability of the LES-based approach is quite sensitive to the acoustic boundary conditions and heat transfer, which are often uncertain. Moreover, this approach requires excessive CPU time [11].

In the forced mode strategy, coupling between the velocity perturbation and the heat release rate perturbation can be expressed by the flame transfer function (FTF) or the flame describing function (FDF) [14–25]. Recently, perturbations of equivalence ratio are also accounted for obtaining FTF or FDF [11,25]. In this strategy, by imposing external forcing at the inlet in experiments or simulations, the gain index and time delay are measured or calculated between the reference point and the flame zone. To avoid unnecessary reflected waves, a non-reflecting boundary condition must be specified [1]. Compared to the self-excited mode strategy, this method requires much less computational cost if FTF or FDF data are available. Due to this advantage, this forced mode strategy has gained popularity for the analysis of thermoacoustic instability over a wide range of the operating conditions and various geometric conditions [19,21–24]. However, in highly turbulent swirling flames, it is difficult to obtain accurate FTF or FDF data. Depending upon FTF or FDF models, reference points, and perturbation ratios, the values of FTF or FDF could be quite different, even at the same frequency. Even if the difference in the values of FTF or FDF is

* Corresponding author.
E-mail address: ymkim@hanyang.ac.kr (Y. Kim).

http://dx.doi.org/10.1016/j.applthermaleng.2016.06.143
1359-4311/© 2016 Elsevier Ltd. All rights reserved.

Nomenclature

Roman symbols

a	hole radius
c	speed of sound
d	distance b/w holes
f	frequency
f_{tar}	target frequency
h	plate thickness
k	wave number
l_{eff}	effective length (=πa/2 + h)
n	gain index
p	pressure
Q	heat release rate
t	time
U	velocity
ω	angular frequency
A	area of the hole ($=\pi a^2/2$)
I_1	modified Bessel functions of the first kind
K_0	$=\pi a^2/l_{eff}$
K_1	modified Bessel functions of the second kind
K_R	Rayleigh conductivity
L_c	back cavity length
P	porosity
Q	volume flow rate
R_p	reflection coefficient of the plate
U_{hole}	velocity in the hole of the plate
He	Helmholtz number of the back cavity ($=kL_c$)
St	Strouhal number ($=wa/U_{hole}$)
St_l	Strouhal number of an effective length ($=wl_{eff}/U_{hole}$)
FDF	flame describing function
FTF	flame transfer function
$\Re e[\ ]$	real component
Im[ ]	imaginary component

Greek symbols

α	internal damping rate
α'	effective damping rate
γ	heat capacity ratio
ρ	density
σ	contraction ratio (=0.75)
τ	time delay
ζ	impedance

Subscripts

$^-$	mean part
$'$	fluctuation part
$^\wedge$	complex value in a frequency domain
$_i$	imaginary part
$_r$	real part
$_{ref}$	reference point
$^+$	positive direction
$^-$	negative direction
$_1$	back cavity
$_2$	plenum
$_3$	chamber
$_p$	perforated plate

small, it potentially results in somewhat different resonant frequencies or growth rates. Thus, special care is required to obtain the values of FTF or FDF. However, the FTF-based approach is only applicable for analyzing linear combustion instability. Alternatively, the FDF-based approach can overcome the shortcomings of the FTF-based approach. In the FDF-based approach, the eigenfrequencies are varied by increasing the velocity perturbation ratios [14,20,22]. This unique feature enables the FDF-based approach to accurately predict the resonant frequencies and nonlinear combustion dynamics such as the hysteresis, frequency shifting, mode change, or limit-cycle instability [14,18–21].

To stabilize the oscillation arising from the combustion instability, these three kinds of damping devices are mostly used: quarter-wave resonators, Helmholtz resonators, and perforated plates [26]. Quarter-wave resonators are often used to suppress transverse high-frequency unstable modes while Helmholtz resonators are typically used to attenuate low-frequency unstable modes. However, Helmholtz resonators and quarter-wave resonators have a limited ability to stabilize unstable modes only in a narrow bandwidth of frequency. Thus, if combustion instability exist in a several unstable modes in a system, above two kinds of dampers should be installed with tuning to each resonant frequency. On the other hand, perforated plates overcome this drawback of these two kinds of dampers, so it can be a promising alternative to other stabilization devices, especially in gas turbine combustors [26–31]. Up to now, there are not many researches in effectively designing and using perforated plates to damp the thermoacoustic instability. A state-of-the-art method is using the Howe model when designing perforated plates [26,28–30], which has the shortcoming of being unable to account for the plate thickness. In addition, during the obtaining design parameters, several simplification processes are required.

In order to solve these weaknesses, the modified Howe model is even utilized [26,30]. Nevertheless, same weaknesses still exist in this approach because of the simplification processes. To design dampers with the original or modified Howe model, it is assumed that dampers are operated only at very large or very small Strouhal numbers [26,30]. If dampers need to be operated at some moderate Strouhal numbers due to technical constraints, it is quite difficult to find the optimal attenuation conditions of the dampers. Thus, in order to overcome these drawbacks of the Howe model, the present study has proposed the new approach which adopts the Luong model [31] based on a modified form of Cummings' empirical equation. Compared to the conventional approaches [26,30], the present method utilizing the Luong model has the basic advantages which allows one to design perforated plates over the whole range of Strouhal numbers. Moreover, in contrast to the previous method using the Howe model, this approach can fully take account for the effects of plate thickness and all design parameters of the dampers can be obtained without any simplification processes.

In the present study, the FDF-based approach has been applied to analyze the thermoacoustic instability in a lab-scale lean premixed combustor [22,23]. Computations are carried out for a wide range of combustor and plenum lengths. Based on the numerical results, instability maps of the combustor are generated. The numerical results are compared with the experimental data in terms of the eigenfrequency and limit-cycle velocity perturbation ratios. Finally, the present study also attempted to find the effective stabilization conditions for the perforated plate and numerically evaluated the damping effects of the perforated plate.

2. Theory of numerical models

2.1. Helmholtz equation

By assuming linear acoustics and isentropic variation, the wave equation can be derived from the Navier-Stokes equation. It is also

assumed that fluctuations in the pressure, density, and velocity are much smaller compared to the mean values: $p' \ll \bar{p},\ \rho' \ll \bar{\rho},\ u' \ll \bar{c}$. In linear acoustics, the instantaneous flow variables are decomposed as the mean value and perturbation. In this derivation, viscous forces and the mean flow speed are neglected, and the speed of sound is used as the reference speed. With these assumptions, the well-known wave equation can be derived from the conservation equations of mass, momentum, and energy in the presence of a heat source term.

$$\frac{1}{\bar{c}^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \frac{\gamma - 1}{\bar{c}^2}\frac{\partial q'}{\partial t} \tag{1}$$

Here, γ, t, and q are the specific heat ratio, time, and heat release, respectively. The fluctuating term of the heat release rate represents the combustion effects on the acoustics. In the framework of linear acoustics, fluctuations can be formulated in the form of a harmonic wave. Thus, temporal and spatial variations are decomposed as:

$$p'(\vec{x}, t) = \Re e[\hat{p}(\vec{x})e^{-i\omega t}] = e^{\omega_i t}\Re e[\hat{p}(\vec{x})e^{-i\omega_r t}] \tag{2}$$

Here, $\Re e[\Phi]$ denotes the real part of the complex variable Φ, and ω_r and ω_i are the real and imaginary components of the angular frequency ω, respectively. Using the harmonic variation for the fluctuating variables, the wave equation in the time domain becomes the following Helmholtz equation in the frequency domain:

$$k^2\hat{p} + \nabla^2\hat{p} = \frac{i\omega\hat{q}(\gamma - 1)}{\bar{c}^2} \tag{3}$$

where k is the wave number and $\hat{q}$ can be modeled by FTF or FDF.

2.2. Flame transfer function

It is crucial to model a heat release rate perturbation term in the Helmholtz equation. Crocco devised the flame response function of compact flames [32] to relate the heat release rate perturbation term with the velocity perturbation term. A slightly modified Crocco's model has been widely utilized for analyzing thermoacoustic instability [1,2]. In the frequency domain, FTF is expressed as:

$$FTF(w) = \frac{\hat{q}/\bar{q}}{\hat{u}_{ref}/\bar{u}_{ref}} = n(w)e^{-i\omega\tau(w)} \tag{4}$$

Here, u_{ref} is the velocity measured at the reference point. For thin flames, FTF is a function of only frequency and is determined by the gain index n and the time delay τ. In a real situation, however, FTF depends on flame shapes as well as operating conditions. Thus, each position in the flames has a different gain index or time delay [8,24]. Since calculating FTF in the computational nodes of the whole flame zone is time-consuming, the flame regions are often divided into several sections [24]. It is also necessary to note that the accuracy of FTF is considerably influenced by the choice of the reference point where the velocity fluctuations are obtained [33]. Thus, the reference location should be chosen such that it is sufficiently close to the flame zone [33]. In addition, FTF should cover the whole frequency range where instability can occur. For this reason, prior to the FTF-based instability analysis, the Helmholtz equation with the spatial variation of the temperature (without FTF) is typically solved to roughly estimate the resonant frequencies [2].

2.3. Flame describing function

In actual combustor chambers, thermoacoustic fluctuations frequently occur in a nonlinear zone. However, FTF is calculated when velocity fluctuations are small enough to be in a linear zone. Since the FTF-based approach is unable to analyze nonlinear behaviors, such as frequency shifting, mode change, or hysteresis [14,18–21], Noiray et al. [20,21] proposed the flame describing function to analyze these nonlinear phenomena.

$$FDF(\hat{u}/\bar{u}, w) = \frac{\hat{q}/\bar{q}}{\hat{u}_{ref}/\bar{u}_{ref}} = n(\hat{u}/\bar{u}, w)e^{-i\omega\tau(\hat{u}/\bar{u}, w)} \tag{5}$$

As expressed in the above relation, FDF is simply a group of FTF according to certain ranges of the velocity perturbation ratios. Since the gain index and time delay are dependent on the amplitude of the velocity perturbations, the FDF-based approach essentially has the ability to model the nonlinearities of combustion instability. By increasing the velocity perturbation ratio, the gain index decreases and the phase evolution becomes substantially modified; however, unlike the gain index, this is not straightforward [14,22]. Since FDF accounts for variations of the velocity perturbation ratios, it is clearly able to describe the development of the nonlinear behavior of combustion instability [14,18–21]. Therefore, the FDF-based approach is capable of analyzing the nonlinear combustion instability and the evolution of eigenfrequencies, which cannot be described by the FTF-based approach. Due to the addition of one more variable, the computational cost is squared, but much more information can be obtained for analyzing the thermoacoustic instability.

2.4. Acoustic response of perforated plates

Howe [34,35] theoretically modeled the acoustic response behavior of perforated plates with bias flow. For the acoustic condition, as displayed in Fig. 1, it is possible to find an expression for the reflection coefficient of a perforated plate backed by a resonant cavity that is traversed by a mean bias flow and submitted to normal incident sound waves.

For the low-frequency conditions of acoustic perturbations, the corresponding wavelength is much larger than the hole radius, and the acoustic oscillations of pressure and velocity can be expressed as follows:

In the region of $-L_c \leqslant z \leqslant 0$:

$$\begin{aligned} \hat{p}_1(z) &= p_1^+ \exp(ik(z + L_c)) + p_1^- \exp(-ik(z + L_c)) \\ \bar{\rho}\bar{c}\hat{u}_1(z) &= p_1^+ \exp(ik(z + L_c)) - p_1^- \exp(-ik(z + L_c)) \end{aligned} \tag{6}$$

In the region of $z \geqslant 0$:

$$\begin{aligned} \hat{p}_2(z) &= p_2^+ \exp(ikz) + p_2^- \exp(-ikz) \\ \bar{\rho}\bar{c}\hat{u}_2(z) &= p_2^+ \exp(ikz) - p_2^- \exp(-ikz) \end{aligned} \tag{7}$$

Here, $\bar{\rho}$ is the mean density, $\bar{c}$ is the speed of sound, and k is the wave number. By continuity, the volume flow rate Q has the following relation:

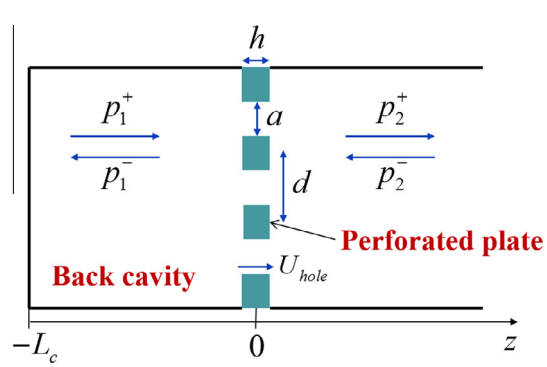

Fig. 1. Schematic of the perforated plate with a back cavity.

$$Q = \hat{u}_1 d^2 = \hat{u}_2 d^2 \tag{8}$$

Here, d is the distance between holes. Howe [34,35] defined the Rayleigh conductivity K_R to characterize the acoustic properties of the bias flow jet.

$$K_R = \frac{-iw\rho_0 Q}{p_2(0) - p_1(0)} \tag{9}$$

The specific impedance ζ of the perforated plate at $z = 0^+$ is expressed as $\zeta(z = 0^+) = \hat{p}_2/(\bar{\rho}\bar{c}\hat{u}_2)$. Combining Eqs. (6), (8), and (9) with the assumption of a full reflecting boundary condition at $z = -L_c$, one can obtain:

$$\zeta_p = i\left[\frac{kd^2}{K_R} - \frac{1}{\tan(He)}\right] \tag{10}$$

Here, $He = kL_c$ is the Helmholtz number of the back cavity and L_c is the back cavity length. The reflection coefficient of a perforated plate can be written as:

$$R_p = \frac{\zeta_p + 1}{\zeta_p - 1} \tag{11}$$

To characterize the specific impedance or reflection coefficient, it is required to model the Rayleigh conductivity. Howe [34,35] developed a model for perforated plates with the assumptions of an infinitely thin thickness, a high Reynolds number in the hole, and low porosity. When the cylindrical vortex sheet is shed with constant velocity U_{hole} at the hole, and also its size is equal to the hole radius a, the Rayleigh conductivity for a perforated plate can be expressed as:

$$K_R = 2a(\gamma - i\delta) \tag{12}$$

Here, γ and δ are determined by the following formulation:

$$\gamma - i\delta = \frac{(\pi/2)I_1(St)\exp(-St) - iK_1(St)\sinh(St)}{St[(\pi/2)I_1(St)\exp(-St) + iK_1(St)\cosh(St)]} \tag{13}$$

where $St = wa/U_{hole}$ is the Strouhal number. I_1 and K_1 denote the modified Bessel functions of the first and second kinds, respectively. To account for the finite thickness of the perforated plates, a modified version of the Howe model was suggested by Jing and Sun [36]:

$$K_R = 2a\left(\frac{1}{\gamma - i\delta} + \frac{2h}{\pi a}\right)^{-1} \tag{14}$$

According to the previous studies [34–36], the attenuation characteristics of the perforated plates can be analyzed with the Howe model or the modified Howe model. However, it is not possible to obtain the accurate values of design parameters of perforated plates by using these two models. To circumvent these difficulties, all of the previous works analyzed for operating conditions with a very low Strouhal number ($St \ll 1$) or a very high Strouhal number ($St \gg 1$). From these limited conditions, the simplified explicit expressions for Eq. (13) can be obtained:

$$\gamma - i\delta = \frac{St^2}{3} - i\frac{\pi St}{4} \text{ when St} \ll 1 \tag{15}$$

$$\gamma - i\delta = 1 - \frac{i}{St} \text{ when St} \gg 1 \tag{16}$$

It is widely recognized that a low-Strouhal number condition yields a much wider absorption bandwidth [26,30] and much better attenuation characteristics at the perforated plate, as compared to the high-Strouhal number condition. However, if dampers are operated at moderate Strouhal numbers (due to technological constraints or high-frequency unstable modes), it is not possible to design dampers accurately. Instead of using the Howe model or modified Howe model, the present study has adopted Luong model, which is based on a modified form of Cummings' empirical equation [31]. In addition, in contrast to Howe model, this approach yields a good approximation for both the nonlinear regime as well as the linear regime. It has been reported that the nonlinear effects associated with the mean velocity can lead to a small reduction in the steady bias flow velocity [31]. When using Luong model, it is possible to design perforated plates that can operate in the whole range of Strouhal numbers. Moreover, compared to the conventional methods [34–36], it is much easier to account for the thickness effects of the plates. All of the optimized design parameters can be easily obtained without further simplification. The formulation of Luong model is expressed below:

$$K_R = \frac{2aSt}{St + 2i/\pi a^2} \quad \text{: for infinitely thin plates.} \tag{17}$$

$$K_R = \frac{K_0 St_l}{St_l + i/\sigma^2} \quad \text{: for finite thick plates.} \tag{18}$$

Here, $St_l = wl_{eff}/U_{hole}$ is the Strouhal number for an effective length, a contraction ratio $\sigma = 0.75$ is suggested, and $K_0 = \pi a^2/l_{eff}$ and $l_{eff} = \pi a/2 + h$ are defined. By combining Eqs. (10) and (18), the expression of the specific impedance of perforated plates can be written as:

$$\zeta_p = -\frac{U_1}{c\sigma^2 P^2} + i\left(\frac{kd^2 l_{eff}}{A} - \frac{1}{\tan(He)}\right) \tag{19}$$

Here, P and U_1 are the porosity and the bulk velocity in the back cavity, respectively.

2.5. Optimal attenuation conditions of perforated plates

The ideal situation occurs when a non-reflective boundary condition is imposed at a perforated plate in a combustor. To meet this requirement, the specific impedance should be equal to −1 [26,30]:

$$\Re e[\zeta] = -\frac{U_1}{c\sigma^2 P^2} = -1 \tag{20}$$

$$\mathrm{Im}[\zeta] = i\left(\frac{kd^2 l_{eff}}{A} - \frac{1}{\tan(He)}\right) = 0 \tag{21}$$

Here, $\Re e[\zeta]$ and $\mathrm{Im}[\zeta]$ indicate the real and imaginary components of the specific impedance, respectively. The porosity P is determined from Eq. (20) and the back cavity length L_c is obtained from Eq. (21):

$$P = \sqrt{\frac{U_1}{c\sigma^2}} \tag{22}$$

$$L_c = \frac{1}{k}\tan^{-1}\left(\frac{P}{kl_{eff}}\right) \tag{23}$$

Thus, the corresponding hole radius can be obtained as $a = \frac{StU_{hole}}{Pw}$. Evidently, by decreasing the Strouhal number St, the performance of the perforated plates is improved [26,30]. Finally, from the definition of the porosity, the distance between holes is given as $d = a\sqrt{\frac{\pi}{P}}$. When the target frequency is chosen, the wave number k is also determined; these four design parameters (P, L_c, a, and d) can be fixed to identify the effective attenuation conditions of the perforated plates.

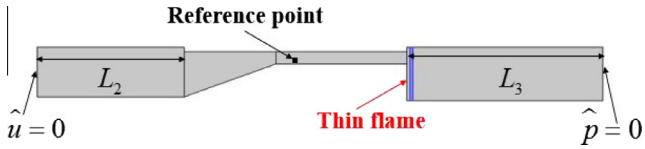

Fig. 2. Modeled combustor configuration and acoustic boundary conditions.

3. Combustor geometry and operating conditions

The precise geometry of the combustor and its flame images are available in the previous studies [22,23]. Fig. 2 shows the axisymmetric combustor geometry, the modeled thin flame, and the acoustic boundary condition at inlet and outlet. The position of the reference point is estimated from the literature [22,23]. The velocity perturbations evaluated at the reference point are used to evaluate the source term of the Helmholtz equation. In computations, idealized acoustic boundary conditions are imposed at the boundaries (inlet : $\hat{u} = 0$ and outlet : $\hat{p} = 0$). The premixed flame condition in this lean-premixed combustor is based on the methane-air mixture with the fixed equivalence ratio, 0.7 and the thermal load, 1.94 kW. The velocity at the reference point is 2.67 m/s. The Helmholtz equation is solved by the finite element method mainly due to its unique strength and robustness to treat the acoustic boundary conditions.

Computations are made for twelve cases, covering a wide range of plenum and chamber lengths, as given in Table 1. Thermodynamic properties for the unburned and burnt gases are also listed in Table 2. To account for the realizable acoustic boundary condition, an end correction is utilized at the outlet ($l^* = l + 0.61R_3$) [37,38]. Here, l and R_3 are the length and radius of the combustion chamber, respectively. Due to the assumption of compact flames, the combustion instability analysis in conjunction with the present FDF approach is based on the global gain index and time delay. In measurements [22], the FDF data were obtained for six different velocity perturbation ratios in the given flame conditions.

The FDF data of Flame A are well documented in the previous work of Silva et al. [22]; those FDF data are available for the frequency range from 0 to 400 Hz. By increasing the velocity perturbation ratio, the gain index is decreased because the energy loss in the system linearly increases over the whole regime. However, the energy gain does not increase linearly in the nonlinear zone. This nonlinear behavior is apparent especially for the high frequency regime [14,22]. However, in terms of the phase, the increase of the velocity perturbation ratio does not result in distinctly different trends. Experimental data indicate that, in the very low frequency region, there exists a small bump in the phase of the measured FDF data [22].

4. Attenuation effects of perforated plates on the combustion instability

First, the present FDF-based approach utilizing these measured FDF data has been applied to analyze thermoacoustic instability for 12 different cases of Flame A in the lab-scale premixed combustor. Fig. 3 shows the computed eigenfrequency trajectories for these 12 cases. Numerical results indicate that the first longitudinal modes

Table 2
Thermodynamic properties for the unburned and burnt gases.

	Temperature [K]	Pressure [atm]	Density [kg/m^3]	Heat capacity ratio
Unburnt gas	300	1	1.177	1.4
Burnt gas	1600	1	0.221	1.4

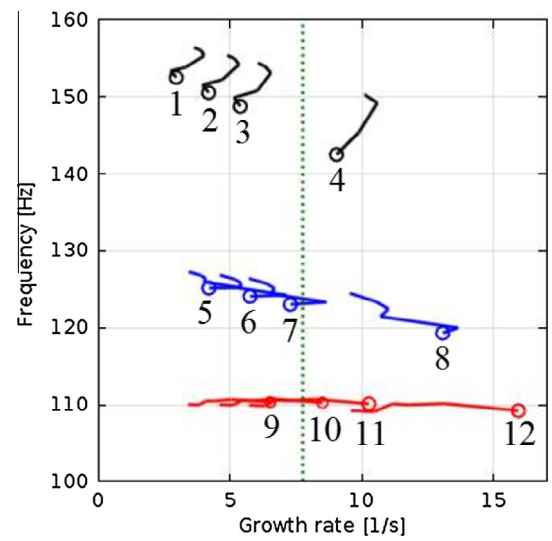

Fig. 3. Eigenfrequency trajectories of the 12 cases.

for the 12 cases are unstable. The vertical dotted line represents the internal damping rate of the system, which is estimated from the following relation [23]:

$$\alpha = \alpha' + w_i \tag{24}$$

Here, α denotes the internal damping rate, and α' and w_i are the measured effective damping rate [23] and the growth rate coming from the numerical calculations, respectively. Similar to the previous study [22], the present approach uses the measured effective damping ratio α' for case 5. Alternatively, in terms of the growth rate, this approach uses the numerically calculated value for case 5 whereas the previous work [22] utilized the calculated growth rate evaluated by averaging the growth rate for all stable cases (including case 5). Due to this different treatment, the computed internal damping rate could be slightly different from those determined in previous works [22,23]. In addition, certain discrepancies in our measurements could be partly attributed to the simplified geometry of the combustor as well as to the slightly different position of the reference point associated with the velocity fluctuations. Circles of each trajectory line represent the starting points of the trajectories ($\hat{u}/\bar{u} = 0.07$). If the growth rate exceeds the internal damping rate of the system, the system becomes unstable. By increasing the velocity perturbations, cases 1–3 are kept stable and case 4 is always unstable. However, this does not imply that the wave oscillations in case 4 will grow continuously. Thus, it is expected that the growth rate for case 4 will be decreased, especially at significantly higher velocity perturbation ratios. For case

Table 1
Dimensions of the combustor for 12 cases.

Case	1	2	3	4	5	6	7	8	9	10	11	12
Plenum length [mm]	96	96	96	96	160	160	160	160	224	224	224	224
Chamber length [mm]	100	150	200	400	100	150	200	400	100	150	200	400

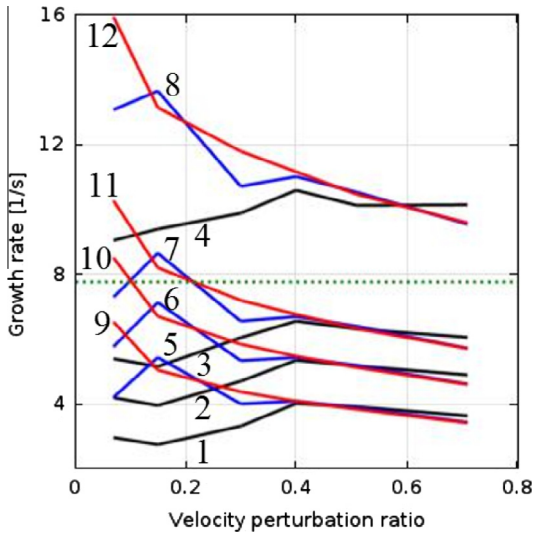

Fig. 4. Growth rates of the 12 cases.

Table 3
Comparison of the numerical results with the experimental data at the limit cycle.

Case	Flame A			
	Resonant frequency [Hz]		Limit cycle velocity perturbation ratio	
	Exp.	Num.	Exp.	Num.
4	137	150 ↑	0.8	0.7 ↑
7	120	125	0.3	0.2
8	116	124	0.9	≒ 0.9
11	115	110	0.4	0.2
12	102	110	0.9	≒ 0.9

4, the computational results agree reasonably well with the experimental data [22]. It can be seen that case 5 and case 6 are always stable, while case 8 is always unstable. The numerical results for cases 5, 6, and 8 agree well with the experimental results. However, for case 7, some slight deviations exist. As displayed in Fig. 4, case 7 is unstable until the velocity perturbation ratio reaches about 0.2, at which point it becomes stable for velocity perturbation ratios higher than 0.2. However, in the experiment, the oscillation grows until the velocity perturbation ratio reaches approximately 0.3. Thus, case 7 is associated with a transitional stability regime. In the computational results, cases 9–11 are stable and case 12 is always unstable. The numerical results are quite similar to the experimental results, but slight deviations exist for case 11. For the unstable cases listed in Table 3, a quantitative comparison is made between the numerical results and experimental data in terms of the resonant frequency and the limit-cycle velocity perturbation ratio. Even if slight discrepancies exist, the present FDF-based approach shows an overall agreement with the experimental results. In cases 4, 8, and 12, since the FDF data are only available for velocity perturbation ratios up to 0.7, the analysis cannot be made for higher perturbation ratios. With simple linear extrapolation, however, the limit-cycle velocity perturbation ratios for cases 8 and 12 are estimated to be 0.9, which closely agrees with the experimental results.

Next, in conjunction with the FDF-based approach, the present study attempted to find the effective stabilization conditions of the perforated plate in Flame A conditions. Considering the resonant frequencies of all cases, we set the target frequency as 120 Hz. We also numerically tested the damping effects of the perforated plate. As discussed above, an alternative approach has been proposed to determine the porosity P by using Eq. (22). This equation determines a porosity value of $P \approx 0.0396$ by calculating the bulk velocity in the back cavity (i.e., $U_1 = u_{ref}A_{ref}/A_1 = 2.67 \cdot 22^2/65^2 \approx 0.306$ [m/s]). Then, to obtain the back cavity length L_c, the effective length $l_{eff} = h + \pi a/2$ must be computed. However, there is no equation to determine the plate thickness h. Thus, parametric studies have been conducted to identify the effects of the plate thickness h on the attenuation characteristics. In accordance with the Strouhal number, the hole radius a is evaluated. By conducting parametric studies for a wide range of Strouhal numbers, the optimized value of the hole radius can be determined. Then, with a fixed Strouhal number, the effects of varying the target frequency and the hole radius are numerically investigated. After optimal design parameters are determined, we impose the corresponding acoustic impedance at inlet, which is well described in Fig. 5.

Fig. 6 presents the magnitude of calculated reflection coefficient of the perforated plate for three Strouhal numbers at a fixed plate thickness ($h = 2$ mm) and the target frequency ($f_{tar} = 120$ Hz). This analysis was made to evaluate the performance of the perforated plate as an effective damper. In the present procedure, if the reflection coefficient is equal to zero at the certain frequency, this implies that there is no reflected wave at the perforated plate; this condition corresponds to the most effective stabilization condition. Here, if the target frequency is specified, the Strouhal number is linearly proportional to the hole radius. As indicated in Fig. 6, by decreasing the hole radius, the bandwidth becomes much wider and can be universally applied. These results suggest that the smaller hole radius (corresponding to the smaller Strouhal number) is more efficient for stabilizing the oscillation induced by combustion. It can be also seen that, besides the target frequency, the conditions with a reflection coefficient of zero appear periodically. This feature could be an additional benefit. However, it is still inconvenient for actual applications because the bandwidth at much higher frequencies becomes progressively narrower. Additionally, conditions that lead to a zero-reflection coefficient rarely appear, especially at higher Strouhal numbers.

Fig. 7 illustrates the effects of plate thickness on the reflection coefficient with a fixed hole radius ($a = 5$ mm) and a specified target frequency ($f_{tar} = 120$ Hz). Numerical results indicate that, by decreasing the plate thickness, the bandwidth becomes much wider. Similar to the influence of a smaller hole radius, much

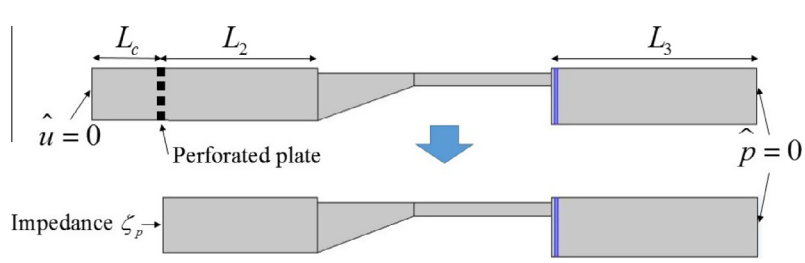

Fig. 5. Computational geometry to consider the effects of a perforated plate in a Helmholtz solver.

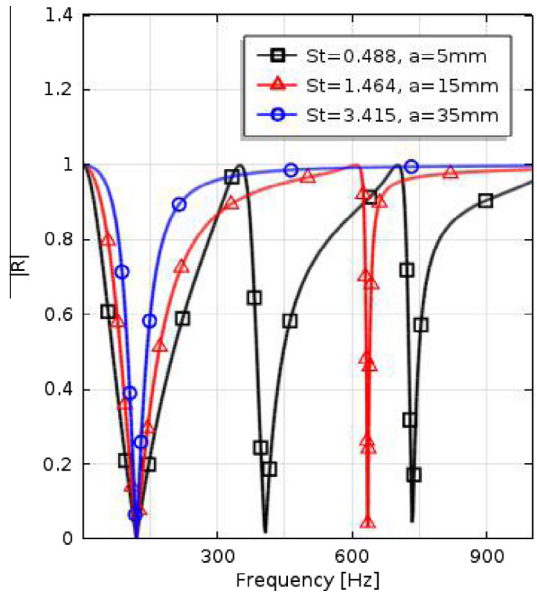

Fig. 6. Magnitude of reflection coefficient of the perforated plate for three Strouhal numbers at a fixed plate thickness ($h = 2$ mm) and target frequency ($f_{tar} = 120$ Hz).

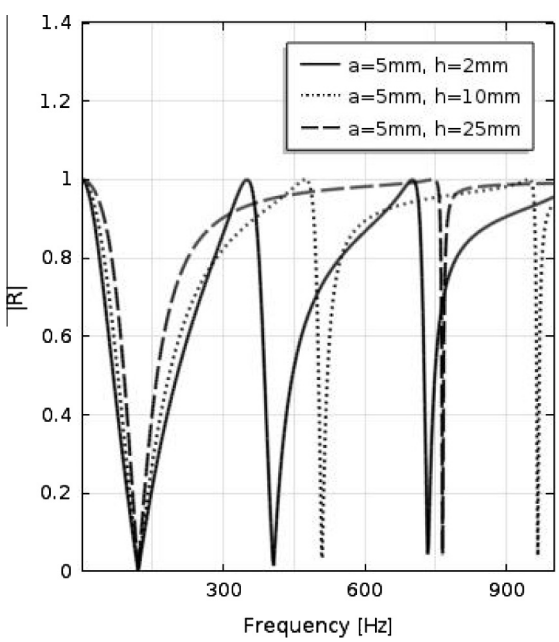

Fig. 7. Reflection coefficients of the perforated plate for three plate thicknesses with a fixed hole radius ($a = 5$ mm) and target frequency ($f_{tar} = 120$ Hz).

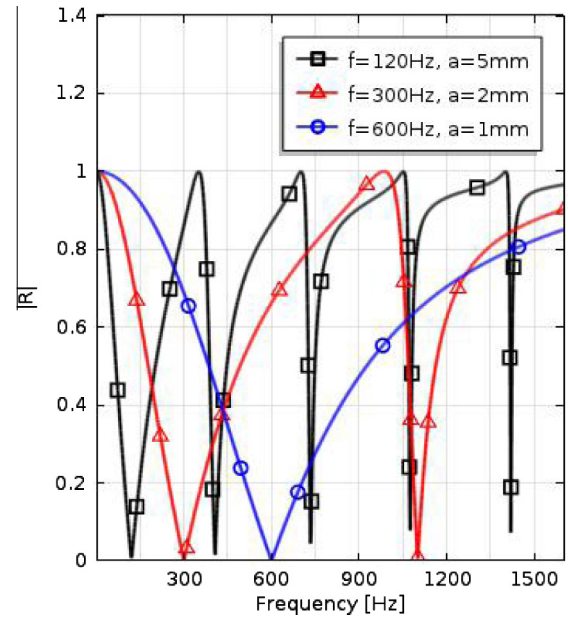

Fig. 8. Reflection coefficients of the perforated plate for three different target frequencies f_{tar} and hole radii a with a fixed Strouhal number ($St = 0.488$).

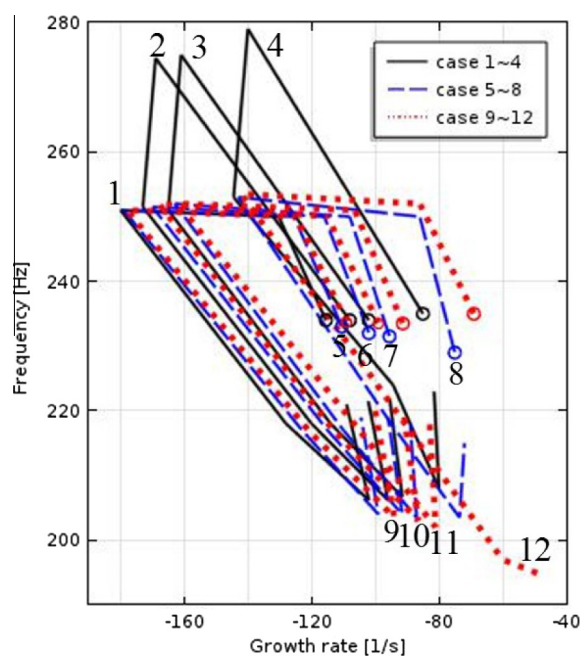

Fig. 9. Eigenfrequency trajectories for the 12 cases attenuated by the perforated plate ($P \approx 0.0396$, $h = 2$ mm, $a = 5$ mm, $f_{tar} = 120$ Hz, and $L_c \simeq 0.4947$ m).

thinner plate thicknesses result in more efficient damping. Thus, a thinner plate is a much better choice for stabilizing oscillations. Here, the periodical appearance of the $|R| = 0$ point and a thinner bandwidth also appeared at higher frequencies.

Fig. 8 shows the calculated reflection coefficients of the perforated plate for three different target frequencies f_{tar} and hole radii a with a fixed Strouhal number ($St = 0.488$). These numerical results reveal that, even with the same Strouhal number, the bandwidth becomes much broader as the hole radius decreases. This trend suggests that, if unstable modes appear sporadically in the higher frequency regime, a perforated plate with a smaller hole radius tuned to a higher target frequency can be very effective at damping all of the unstable modes. It is also observed that the profiles of the reflection coefficient become progressively biased as the frequency increases.

The present parametric studies confirm that a thinner perforated plate with a smaller hole size yields more efficient damping characteristics. Based on the numerical results obtained in this work, effective attenuation conditions for the perforated plate can be estimated as $P \approx 0.0396$, $h = 2$ mm, $a = 5$ mm, $f_{tar} = 120$ Hz, and $L_c \simeq 0.4947$ m. With these optimized design parameters, the corresponding specific impedance can be imposed at the inlet, as opposed to the acoustic boundary condition $\hat{u} = 0$. Under actual conditions, the resonant frequencies can vary from 115 to 160 Hz. This implies that setting the target frequency to

120 Hz is sufficient. In this analysis, the imaginary part of the frequency is not considered while evaluating the target frequency. Fig. 9 shows the calculated eigenfrequency trajectories for the 12 cases attenuated by the perforated plate.

The numerical results clearly indicate that the perforated plate designed in this study has the ability to drastically damp oscillations for all 12 cases. The proposed optimal attenuation conditions for the perforated plate ensure stable conditions for the whole range of velocity perturbation ratios. It can be also seen that these calculated eigenfrequency trajectories show distinctly different damping characteristics, as compared to the undamped results illustrated in Fig. 3. In these numerical results, the resonant frequencies are increased from about 115–160 Hz to 200–280 Hz, while the growth rates are decreased from about 2–16 [1/s] to −40 – −180 [1/s]. For all 12 cases, the first mode of each case was unstable before applying the damper; however, after damping, all of the cases became very stable. Therefore, it has been verified that our proposed passive damper, using a perforated plate, is quite effective for attenuating the combustion instability.

5. Conclusions

The present study focused on analyzing the combustion instability in a lab-scale lean-premixed burner using an FDF-based approach. We also performed the design analysis for the attenuation characteristics of a perforated plate which has the excellent potential for stabilizing the combustion instability encountered in industrial gas turbine combustors. Based on the numerical results, the following summaries can be drawn:

(1) In order to overcome the drawbacks of the conventional approaches utilizing the Howe model, the present study has proposed the new approach employing the Luong model. The model formulation indicates that the present approach has the basic advantages such as the applicability to the whole range of Strouhal numbers and capability to account for the effects of finite plate thickness.
(2) In terms of the resonant frequency and the limit-cycle velocity perturbation ratio, the present FDF-based approach shows an overall agreement with the experimental results even if slight discrepancies exist.
(3) Numerical results indicate that by decreasing the hole radius or the plate thickness, the bandwidth becomes much wider and the applicability could be progressively widened. Even with the same Strouhal number, the bandwidth can become much broader by decreasing the hole radius. It is also found that, if unstable modes appear sporadically in the higher frequency regime, a perforated plate with a smaller hole radius tuned to a much higher target frequency can be very effective at damping all of the unstable modes.
(4) Moreover, the present study also attempted to find the effective stabilization conditions for the perforated plate and numerically evaluated the damping effects of the perforated plate. The perforated plates designed in this numerical study have demonstrated the ability to drastically dampen the oscillations for all test cases. It has been verified that the proposed attenuation conditions of the perforated plate ensure stable conditions over the whole range of velocity perturbation ratios.

Acknowledgements

This work was supported by the Development of 300 MW class Korean IGCC demonstration plant technology of the Korea institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korea government Ministry of Knowledge Economy.

References

[1] T. Poinsot, D. Veynante, Theoritical and Numerical Combustion, third ed., R.T. Edwards, Bordeaux, France, 2011.
[2] P. Salas, Aspects numeriques et physiques des instabilites thermoacoustiques dans les chambres de combustion anulaires, PhD thesis, Uniersite Bordeaux 1.
[3] C.A. Armitage, R. Balachandran, E. Mastorakos, R.S. Cant, Combust. Flame 146 (2006) 419–436.
[4] M. Shahi, J.B.W. Kok, A.K. Pozarlik, J.C.R. Casado, T. Sponfeldner, In ASME Turbo Expo 2013: Turbine Technical Conference and Exposition, Paper GT 2013-94328.
[5] G. Boudier, L.Y.M. Gicquel, T. Poinsot, D. Bissieres, C. Berat, Proc. Combust. Inst. 31 (2007) 3075–3082.
[6] M. Shahi, J.B.W. Kok, J.C.R. Casado, A.K. Pozarlik, Appl. Therm. Eng. 71 (2014) 276–290.
[7] M. Shahi, J.B.W. Kok, J.C.R. Casado, A.K. Pozarlik, Appl. Therm. Eng. 81 (2015) 128–139.
[8] S. Hermeth, Mechnisms Affecting the Dynamic Response of Swirled Flames in Gas Turbines, PhD thesis, Universite den Toulouse.
[9] Benedetta Franzelli, Eleonore Riber, Laurent Y.M. Gicquel, Thierry poinsot, Combust. Flame 159 (2012) 621–637.
[10] S. Roux, G. Lartique, T. Poinsot, U. Meier, C. Berat, Combust. Flame 141 (2005) 40–54.
[11] Pierre Wolf, Gabriel Staffelbach, Laurent Y.M. Gicquel, Jens-Dominik Muller, Thierry poinsot, Combust. Flame 159 (2012) 3398–3413.
[12] F. Proch, A.M. Kempf, Proc. Combust. Inst. 35 (2015) 3337–3345.
[13] G. Lartigue, U. Meier, C. Berat, Appl. Therm. Eng. 24 (2004) 1583–1592.
[14] F. Boudy, Nonlinear dynamics and control analysis of combustion instabilities based on the "Flame Describing Function" (FDF), Other. Ecole Centrale Paris, 2012. English. <NNT: 2012ECAP0056>. <tel-00870770>.
[15] G. Campa, Prediction of the Thermoacoustic Combustion Instability in Gas Turbines, PhD thesis, Politecnico di Bari.
[16] D. Laera, A. Gentile, S.M. Camporeale, E. Bertolotto, L. Rofi, F. Bonzani, In ASME Turbo Expo 2015: Turbine Technical Conference and Exposition, Paper GT2015-42322.
[17] G. Campa, E. Cosatto, S.M. Camporeale, Proceedings of the 2012 COMSOL Conference.
[18] F. Boudy, D. Durox, T. Schuller, G. Jomaas, S. Candel, ASME J. Eng. Gas Turbines Power 133 (2011) P061502.
[19] F. Boudy, D. Durox, T. Schuller, S. Candel, C. R. Mec. 341 (2013) 181–190.
[20] N. Noiray, D. Durox, T. Schuller, S. Candel, Int. J. Aeroacoust. 8 (1) (2009) 157–176.
[21] N. Noiray, D. Durox, T. Schuller, S. Candel, J. Fluid Mech. 615 (2008) 139–167.
[22] C.F. Silva, F. Nicoud, T. Schuller, D. Durox, S. Candel, Combust. Flame 160 (2013) 1743–1754.
[23] P. Palies, D. Durox, T. Schuller, S. Candel, Combust. Flame 158 (2011) 1980–1991.
[24] K.T. Kim, J.G. Lee, B.D. Quay, D.A. Santavicca, Combust. Flame 157 (2010) 1718–1730.
[25] A. Huber, Impact of Fuel Supply Impedance and Fuel Staging on Gas Turbine Combustion Stability, PhD thesis, Technische Universitat Munchen.
[26] A. Scarpato, Linear and nonlinear analysis of the acoustic response of perforated plates traversed by a bias flow, Engineering Sciences [physics]. Ecole Centrale Paris, 2014. English. <NNT: 2014ECAP0038>. <tel-01126834>.
[27] S.H. Lee, J.G. Ih, K.S. Peat, J. Sound Vib. 303 (2007) 741–752.
[28] S. Mendez, J.D. Eldredge, J. Comput. Phys. 228 (13) (2009) 4757–4772.
[29] E. Gullaud, S. Mendez, C. Sensiau, F. Nicoud, T. Poinsot, C.R. Academie, Sci. Mécanique 337 (2009) 406–414.
[30] A. Scarpato, S. Ducruix, T. Schuller, J. Sound Vib. 332 (2013) 4856–4875.
[31] T. Luong, M.S. Howe, R.S. McGowan, J. Fluids Struct. 21 (8) (2005) 769–778.
[32] L. Crocco, J. Am. Rocket Soc. 21 (1951) 163–178.
[33] A. Kaufmann, F. Nicoud, T. Poinsot, Combust. Flame 131 (2002) 371–385.
[34] M.S. Howe, Proc. Roy. Soc. A366 (1979) 205–223.
[35] M.S. Howe, Acoustics of Fluid-Structure Interactions, Cambridge University Press, Cambridge, UK, 1998.
[36] X. Jing, X. Sun, J. Acoust. Soc. Am. 106 (5) (1999) 2436–2441.
[37] H. Levine, J. Schwinger, J. Propul. Power 19 (5) (1948) 383–406.
[38] S. Rienstra, A. Hirschberg, An Introduction to Acoustics, Eindhoven University of Technology.